

Let's learn about **satellites!**

Space mission and **satellite** design course

Satellite auxiliary systems: Thermal and Radiation control

Thermal control

Scope:

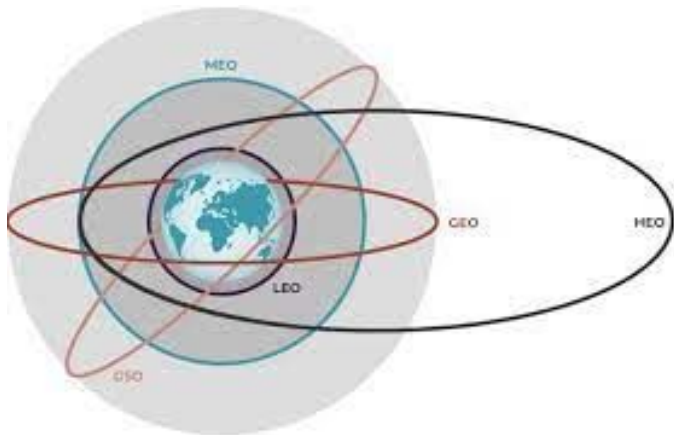
- Keep electronic and mechanical parts temperature inside the operative range
- Minimize thermal deformation

Solution:

- Passive method: surface, painting, radiator, heat pipe, insulating cover
- Active method: heater and cooler

Design requirements

- **Space environment:** mission profile, orbit and internal heat dissipation
- **Components temperature range:** maximum and minimum temperature for operation and non operation



Component	Operating Temp limit (°C)
Emhiser TX	-40 to +85
OBC (mother board)	-40 to +85
SD card	-40 to +85
RTC coin battery	-30 to +80
OBC (processor board)	-40 to +85
EPS	-40 to +85
Battery	-10 to +50
Helium receiver	-30 to +70
Magnetometer	-120 to +50
Sunsensor	-40 to +125
Solar cell	-100 to +100
Structure	-45 to +65

Heat exchange in space

$$Q_e = (\sigma \varepsilon T^4) A \quad [\text{W}] \quad \text{emitted power}$$

- **Radiation:** between surface and space not in contact

$$Q_r = \sigma \epsilon A F (T_2^4 - T_1^4)$$

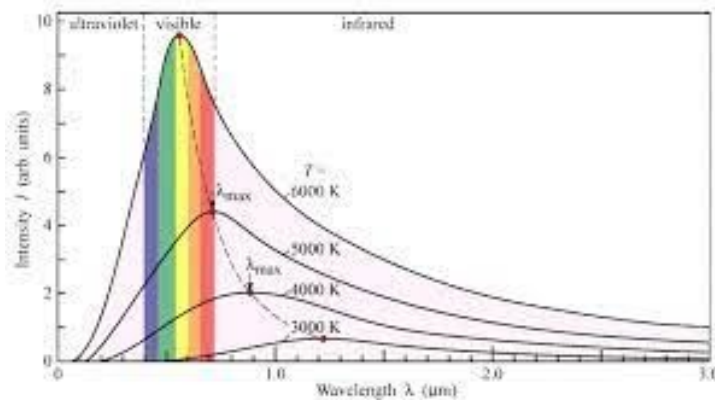
$$\sigma = 5,67 \cdot 10^{-8} [\text{W/m}^2 \text{K}^4] \quad \text{Stefan-Boltzmann constant}$$

ε = energy emitted/**black body** energy ($r=t=0$, $\alpha=0$)

T = body temperature

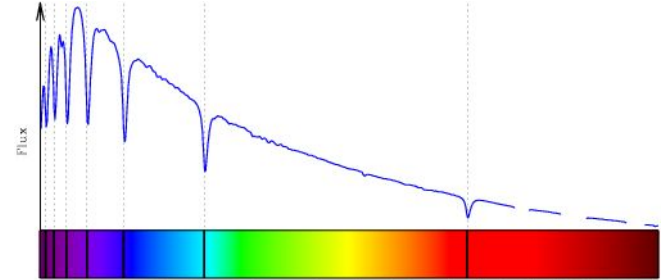
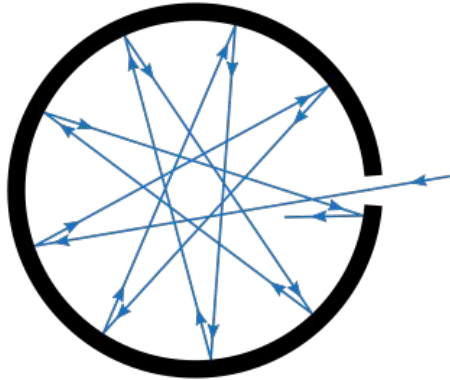
A = emitting surface

F = view factor

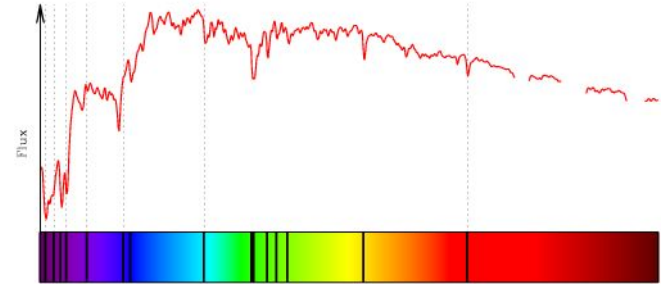


Black body

Idealized real body that absorbs all the incident electromagnetic radiation and the emitted radiation spectrum depends only by the body temperature



A-star spectra show strong hydrogen absorption features (marked with dotted lines).



G-star spectra have weak hydrogen features, and also lines from many other elements.

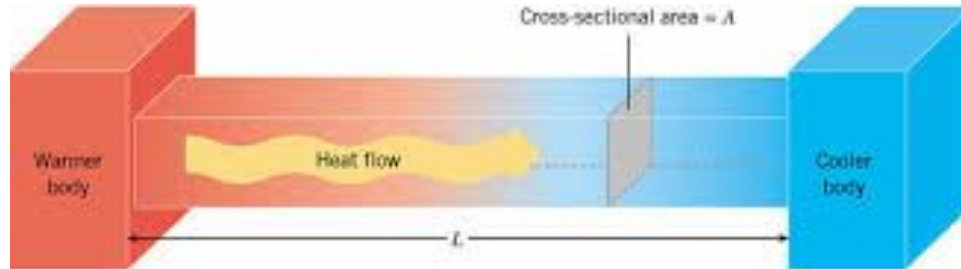
Heat exchange in space

- **Conduction:** between bodies in contact

$$G_c = k A \frac{(T_2 - T_1)}{l}$$

l = distance between bodies

k = thermal conductivity



Specific heat and thermal conductivity

Specific heat capacity [J/Kg K]

amount of heat to add to one unit of substance to increase the temperature of one unit

Thermal conductivity [W/m K]

capacity of material of transport heat

Material	Thermal conductivity at 25°C (W/m K)	Specific heat capacity	
		(J/kg K)	(J/m ³ K)
PA 6	0.25	1600	1824
PLA	0.13	1800	2340
Air	0.024	1005	1231
Wood (oak)	0.17	2000	1833
Steel	43	490	3822
Aluminium	205	870	2484
Water	0.58	4182	4190
Concrete	1.7	880	2122

Heat flux

- Q incident
- Q absorbed
- Q transmitted
- Q reflected
- Q emitted

$$Q_i$$

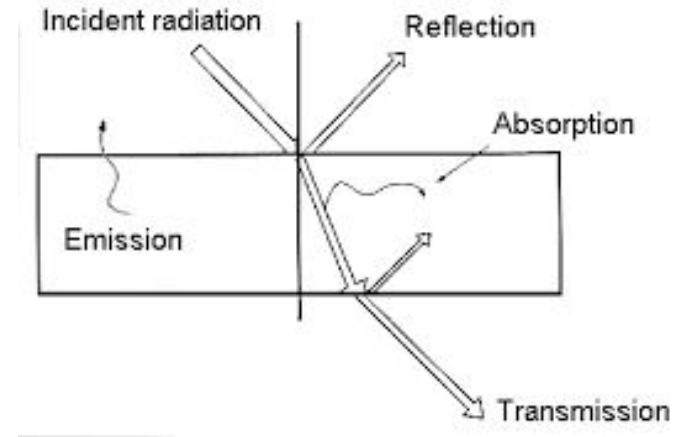
$$Q_a/Q_i = \alpha$$

$$Q_t/Q_i = t$$

$$Q_r/Q_i = r$$

$$Q_e/Q_i = \varepsilon$$

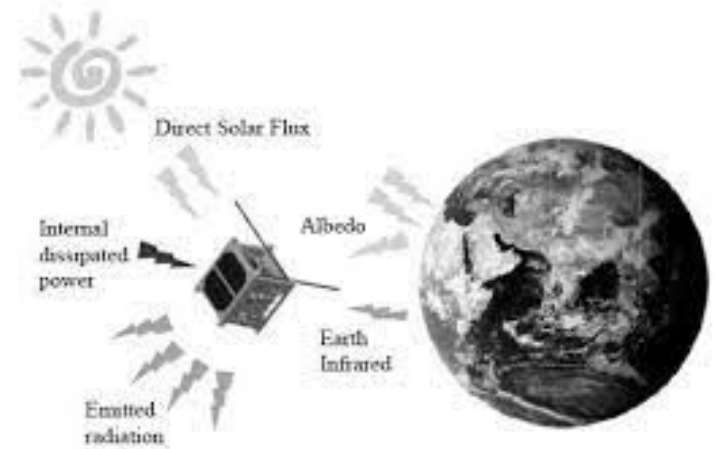
$$\alpha + t + r + \varepsilon = 1$$



Heat source

- Solar radiation
- Albedo radiation
- Infrared radiation
- Internal source
- Emitted radiation

$$\begin{array}{l} Q_s \\ Q_a \\ Q_{ir} \\ Q_i \\ Q_e \end{array}$$



- Hot case
- Cold case

$$Q_s + Q_a + Q_r + Q_i = Q_e$$
$$Q_{ir} = Q_e$$

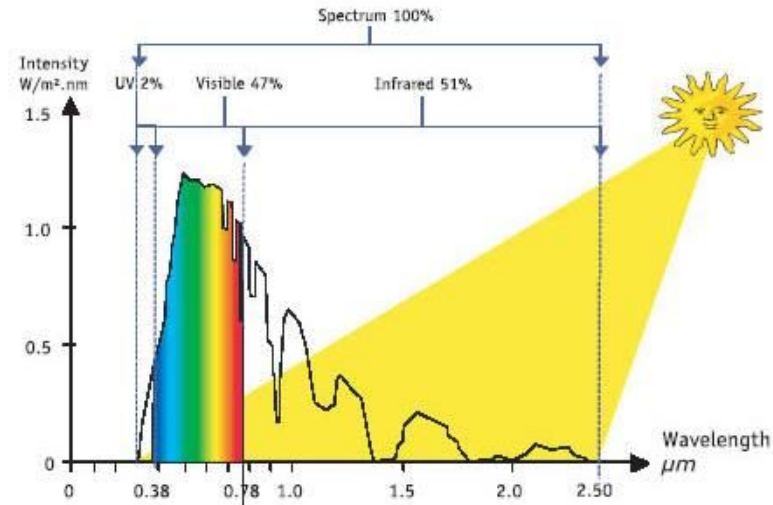
Solar radiation

Radiation emitted by the sun

- Solar spectrum equal to **black body** at 5800 K
- Solar constant $G_s = 1370 \text{ W/m}^2$

$$G_s = (3.85 \cdot 10^{26} \text{ W}) / (4 \pi d^2)$$

$$Q_s = G_s \alpha A$$



Albedo radiation

Solar radiation reflected by the planet

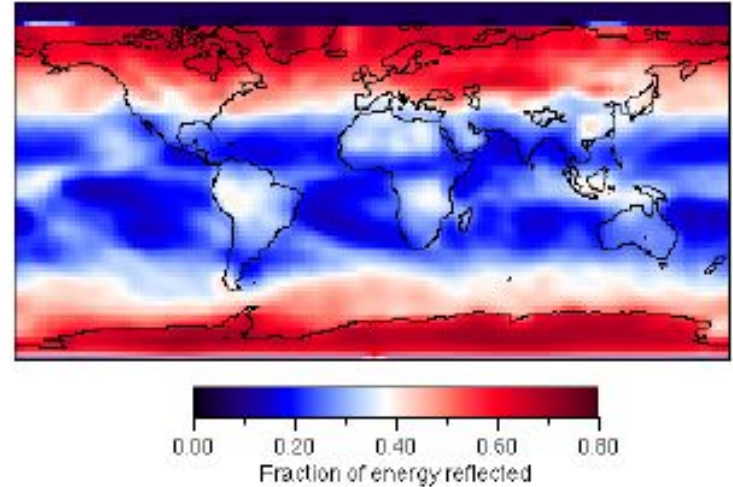
- Change in time due to the reflecting surface:
 - Clouds = 0,8
 - Ice= 0,6
 - Green = 0,03-0,3
 - Mean value = 0,35

$$G_a = G_s \alpha F$$

$$F = \frac{R_E}{(R_E + h)} K_a$$

$$K_a = 0,664 + 0,521r - 0,203r^2$$

$$Q_a = G_a \alpha A$$



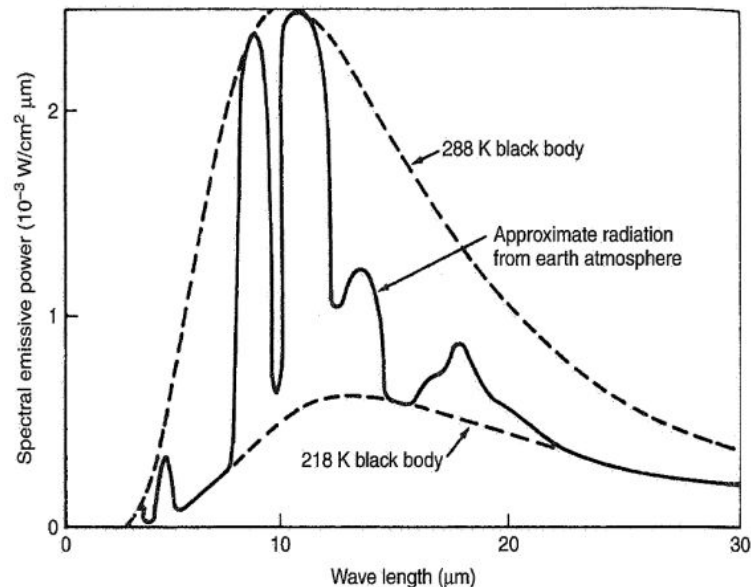
Infrared radiation

Radiation emitted by the Earth

- Earth spectrum equal to **black body** at 288 K
- Earth constant $G_{ir} = 237 \text{ W/m}^2$

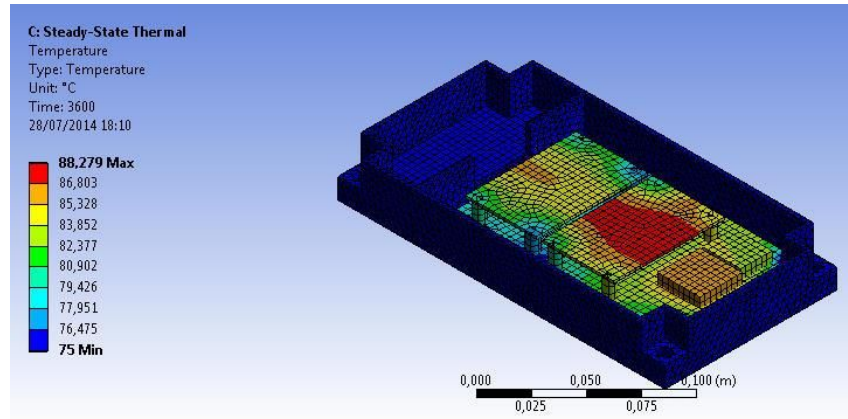
$$G_{ir} = \frac{(237 \text{ W} \cdot R_E^2)}{(R_E h)^2}$$

$$Q_{ir} = G_{ir} \alpha A$$



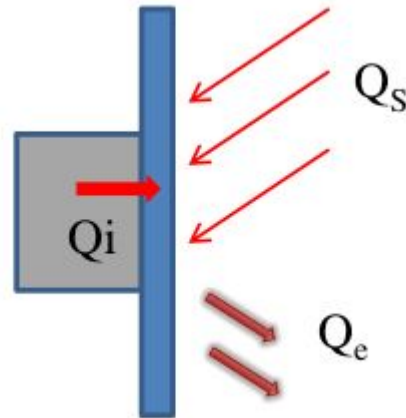
Internal source

- Heat produced by internal electronic system
- Change during phase operations
- Can be increased to protect electronic from freezing



Emitted radiation

- Heat dissipated by the satellite external surface toward deep space
- Can be controlled to dissipate more heat for protect electronic from overheating



Thermal equilibrium

Balance between heat received by internal and external source and heat dissipate to the space

$$G_s \alpha A + G_a \alpha A + G_{ir} \epsilon A + Q_i = \sigma \epsilon T^4 A_{sat}$$

$$T = \left((G_s \alpha A + G_a \alpha A + G_{ir} \epsilon A + Q_i) / (\sigma \epsilon A_{sat}) \right)^{1/4}$$

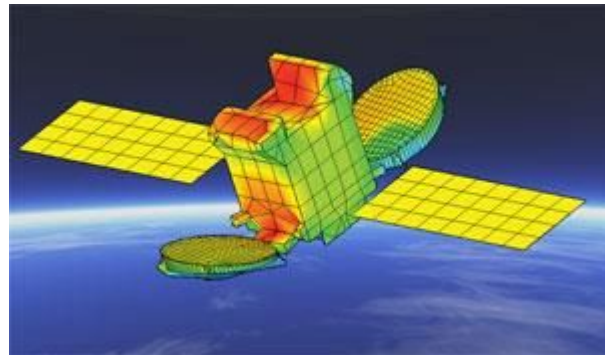
$$T_{min} = \left((G_{ir} \epsilon A + Q_{imin}) / (\sigma \epsilon A_{sat}) \right)^{1/4}$$

$$T_{max} = \left((G_s \alpha A + G_a \alpha A + G_{ir} \epsilon A + Q_{imax}) / (\sigma \epsilon A_{sat}) \right)^{1/4}$$

Thermal analysis

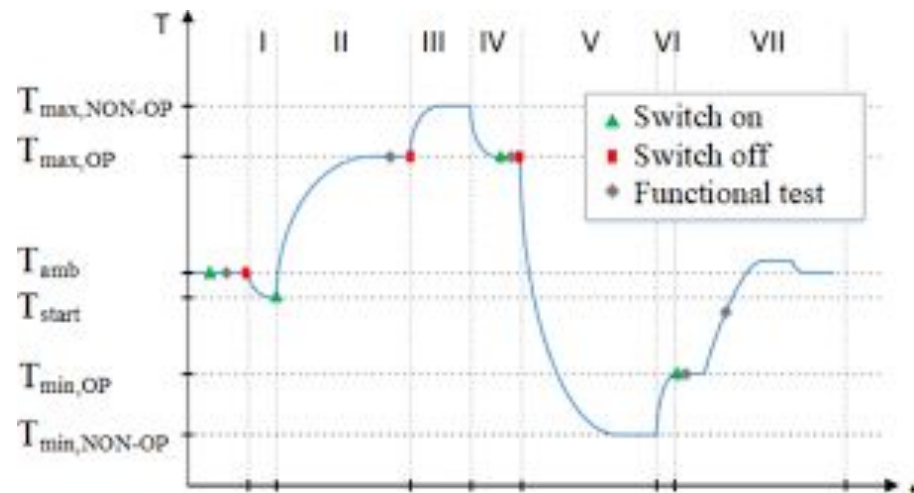
Numerical simulation of a thermal model to estimate the behaviour of the satellite in the real environment

- **Single node:** satellite simplified like a point mass body with all surfaces and material properties
- **Multi nodes:** satellite simplified with main parts related each other by conductive and radiative properties
- **Numerical model:** finite element method analysis



Testing

- Thermal ambient
- **Thermal vacuum**
- Operating temperature
- Non operating temperature
- Functional test



Passive control

Control system are not required

- Surface finish
- Insulating material
- Radiator
- Heat pipe
- Thermal strap

Surface finish

The material on the external surface can be selected for particular absorptivity and emissivity properties: color, material, matte or glossy

**Ageing of the material,
dirty surface**

Surface coating	α	ε	$\frac{\alpha}{\varepsilon}$
White TiO ₂ coatings	0.19	0.94	0.20
White epoxy coatings	0.248	0.924	0.27
White enamel	0.252	0.853	0.30
Polyester film	0.17	0.5	0.34
Aluminum silicon coating	0.25	0.28	0.89
Grey TiO ₂	0.87	0.87	1.00
Black paint	0.975	0.874	1.12
Aluminum based coatings	0.54	0.45	1.20
Aquadag coatings	0.782	0.49	1.60
Aluminum foil	0.192	0.036	5.33
Polishing gold coating	0.301	0.028	10.8

Insulating material



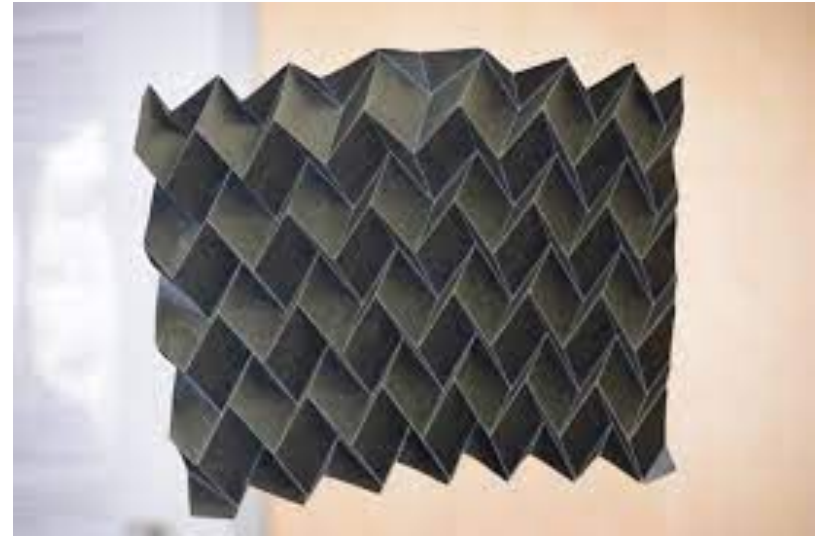
Layers of insulating material separate the component from the environments.

Multi Layer Insulation:
aluminized Mylar layers (low emissivity) separated by nylon (low heat conduction)

If atmosphere is present it's necessary to use **aerogel** or **foam**

Radiator

Heat exchange with large surface to radiate heat in the space: maximum emissivity and minimum absorptivity



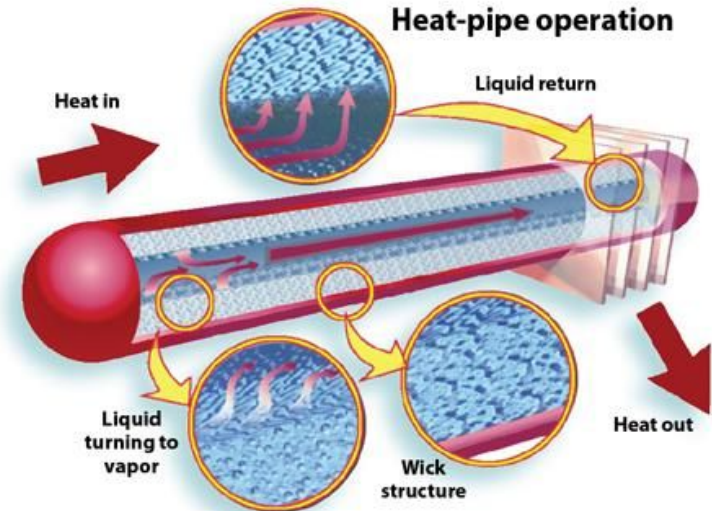
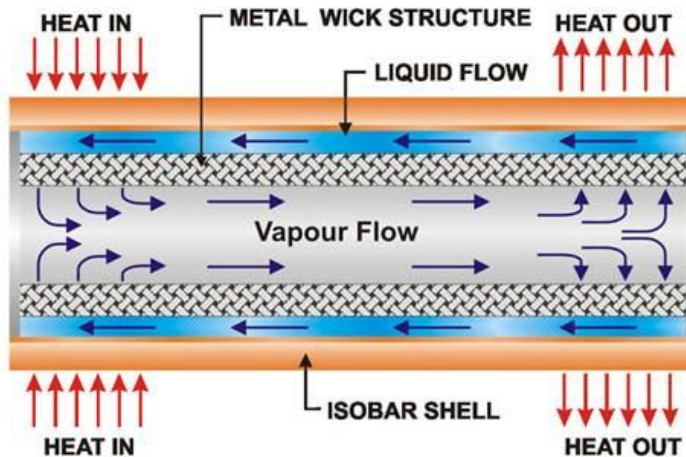
Thermal strap

High thermal conductivity material to create thermal path



Heat pipes

Move heat with high efficiency due to gas-liquid phase change.
200 time more efficient than conduction



Active control

Control system and energy are required.
Heavier and complex

- Heaters
- Liquid loops
- Louvres

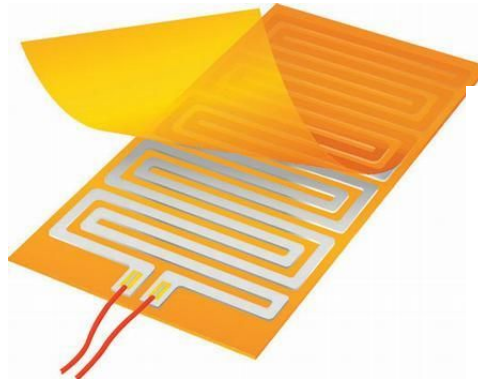
Heater

It increase the temperature with electric power:

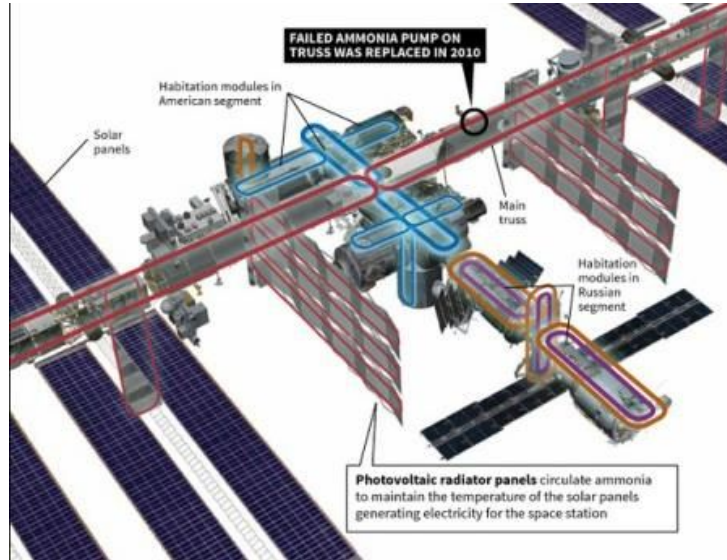
- protection against low temperature
- keep temperature constant
- pre-heating phase for critical component

Patch heater

Cartridge heater



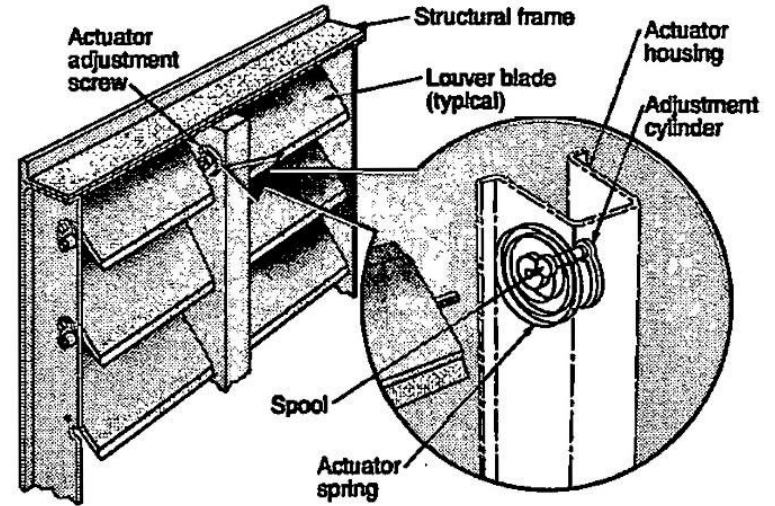
Liquid loops



Complex system where a coolant liquid transport heat inside radiator to cool down electronic components

Louvres

Mechanical device that change the total emitting surface of a radiator with a cover



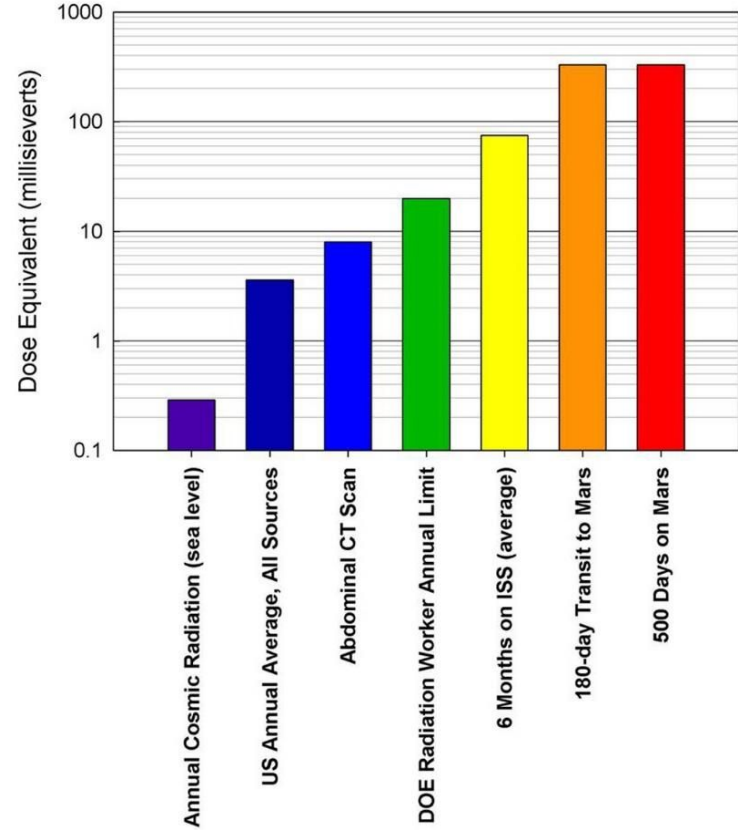
Design process

1. Identify heat sources
2. Identify Cold, Hot and Transitory phase
3. Perform thermal analysis: select thermal control systems
4. Sizing: size and location of thermal control devices
5. Testing

Electromagnetic control

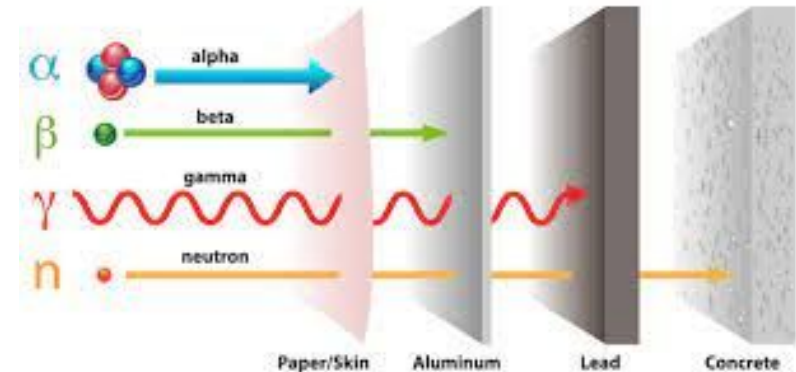
Radiation dose: energy deposited by an energy particle in a component [rad]

1 rad = 0.01 sievert



High energy particle

- **Single-event upset:** change of state in a memory cell but doesn't damage the component
- **Single-event latch-up:** produce an increase of current, it can damage the component
- **Single-event burnout:** complete destruction of the components



Radiation protection

- Radiation hardened
- Shielding
- Redundancy
- Error detection and protection
- Latching circuit

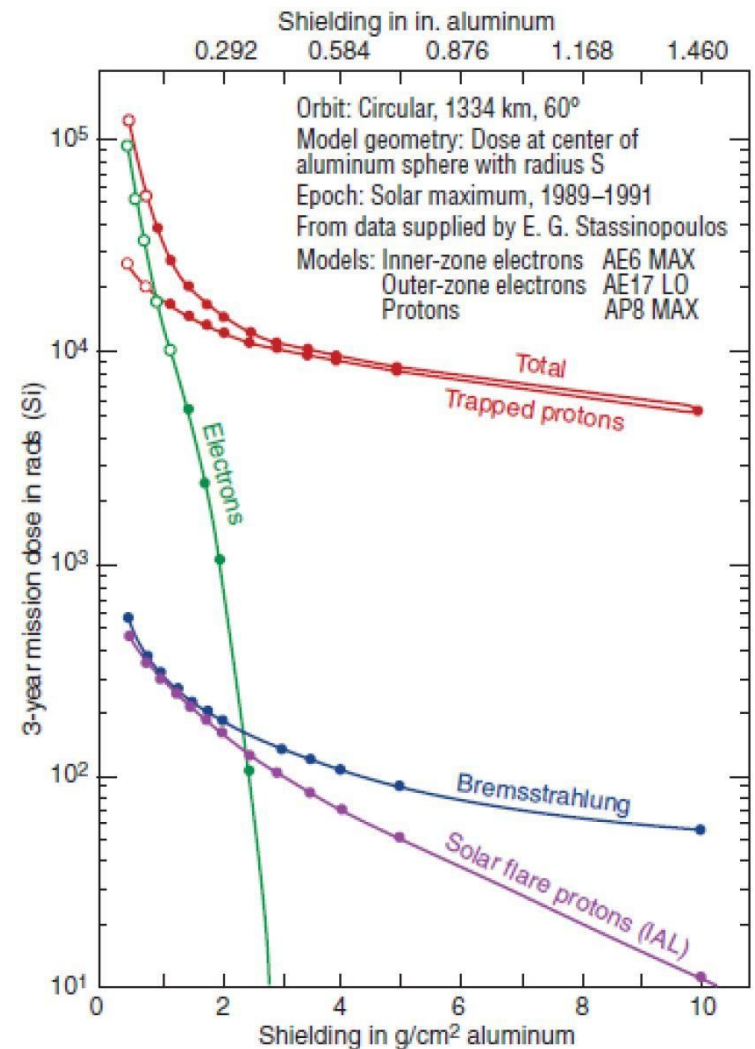
Radiation hardened

Hardened chips are manufactured on insulating substrates instead of the usual semiconductor wafers.

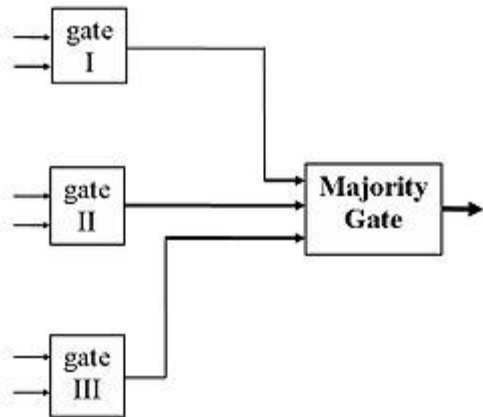


Shielding

Metallic plate like
aluminum or other
metal with high atomic
number like tungsten



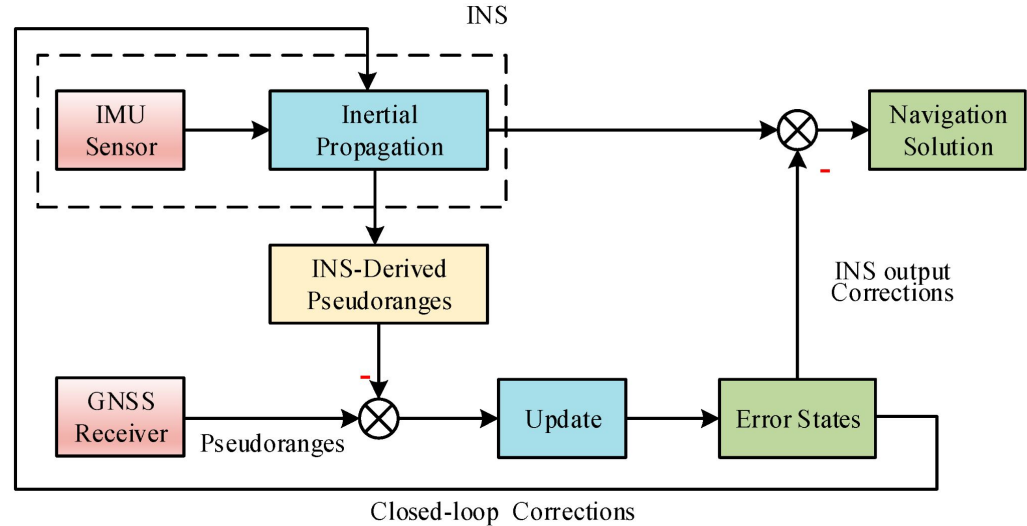
Redundancy



Critical components are redundant:
there are more than one system to do
the same task and if one of them fails
the control module can detect the
error and use the correct result

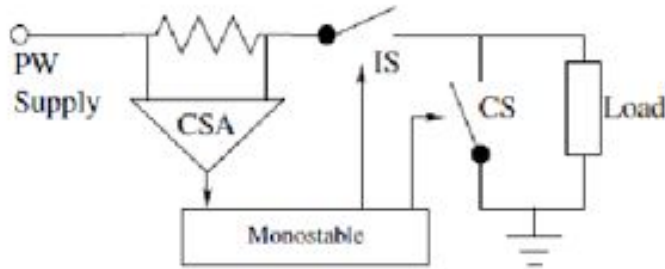
Error detection and protection

The software is able to control the data and recognize if there are anomalies



Latching circuits

Dedicated circuits for the protection of critical system by limiting the current adsorption



Satellite interface

- RBF
- Umbilicals
- Deployment switches
- Electrical ground support equipment EGSE

RBF

Remove Before Flight device that keep all the system turned down during the pre-launch phase



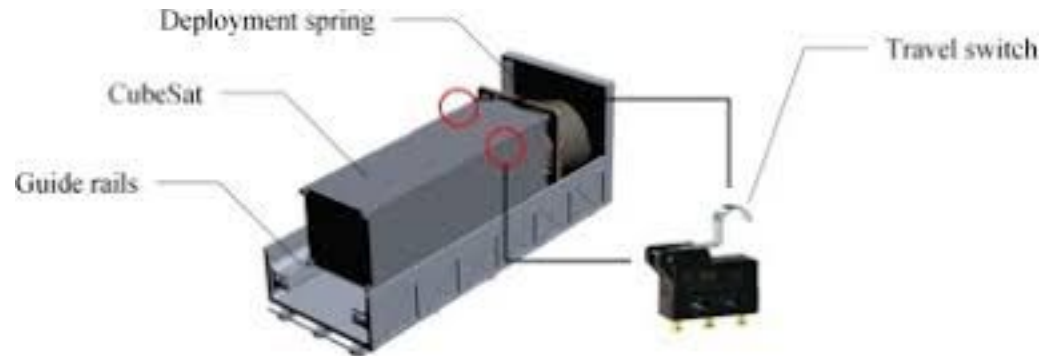
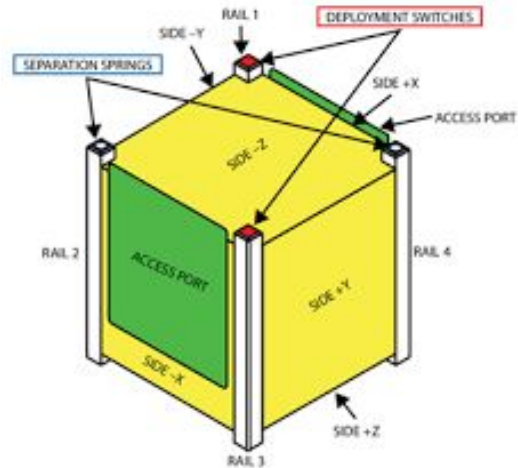
Umbilicals

Accessible connection to the on board computer to allow configuration with the satellite already completely assembled



Deployment switches

These switches are mechanically activated when the satellite is deployed and turn on all the system



EGSE

Electrical Ground Support Equipment

All the specific devices to operate electrical test before flight and simulate different scenarios



Questions?

1. Which are the main heat sources in space?
2. What is a passive heat control?
3. What is an active heat control?
4. Which is the most important thermal test?
5. How can we protect electronic circuits?
6. What is an RBF?